Environmental & Ecological Statistics

Group projects

1 Overview

For the Environmental and Ecological Statistics Group Projects you will take on the role of a statistician consultancy team that will tackle a real-world environmental challenge. For this project, you will be placed into groups and be tasked with writing a concise report summarising the results of a statistical analysis that you have conducted. Your objective is to explore the statistical basis of environmental directives, legislation, and compliance testing while conducting a robust analysis of a given environmental data. You will work in groups of 4 to 5 to produce a professionally structured report summarising your statistical findings. This project will not only enhance your analytical skills but also prepare you for careers in consultancy, research, and policy-making.

2 Project details

Section 4 describes the projects from which each group has been allocated one to work on for this assignment (the group number corresponds to the dataset allocated). Please ensure you are working with the correct dataset and are answering your assigned question of interest.

> **Note**
>
> - The data sets are available to download from Section 4 and on Moodle through the **Group Project Data Sets** link
> - The Group Project Data Set Allocations section contains the information of the classmates you will partner up for the project. Please contact the classmates in your group. There, you will also find out which project your group has been assigned to. The group number corresponds to the dataset your team has been allocated to, e.g., *Group 3* will be analyzing Project 3 - Modelling Sulfur dioxide concentration in California.

2.1 Project Goals

Each group will submit:

1. A self-contained report summarising the main findings of your analysis.
2. A reproducible R code used to produce the results in the report.

This assessment is worth 25% of your final grade for the course and both documents should be submitted by the date stated on the Moodle course page.

- **Analyse the allocated data set using R and store code**

Analyse the data in a collaborative fashion. You can create an R script a quarto or markdown file. If you are familiar with GitHub you can also use it to share your code with your group. You will need to upload a copy of your code (in `.R`, `pdf` or `html` formats) along with your final report.

- **Produce a report on your analysis**

  Your report should summarise the statistical analysis you conducted and a thorough interpretation of your model in relation to the data analysed. You should elect one member of your group to be responsible for submitting the group report as a **.pdf/html** file via the **Group Project Submission** link in section **Group project** on the Data Analysis Moodle page.

  You should submit your group report as `Group_##_Analysis.pdf/html`. Even though one person will be elected to submit the report, all members of the group should be able to see what has been submitted on Moodle.

  Make sure that in your report:

  - all R output, including figures and tables, are appropriately labelled and presented;
  - R code should **not** be included in the body of the report; and
  - the report should be no longer than 6 pages.

  Your group report should include:

  - An appropriate **Title**;
  - An **Introduction** section detailing the data set and question of interest;
  - An **Exploratory Analysis** of the data;
  - A **Formal Analysis** of the data; and
  - Finish with your **Conclusions**.

2.2 Marking scheme

This assessment will be marked out of **18 MARKS** with the report worthing a total of **15 MARKS**, which are broken down as follows:

1. **Title** and **Introduction** `[1 MARK]`

   - Appropriate title

   - Description of background to problem and questions of interest.

2. **Exploratory data analysis** `[5 MARKS]`

   - Appropriate exploration of the allocated dataset using multiple exploratory techniques and identification of possible patterns/anomalies in the data.

   - Appropriate comments relating to the summary statistics / plots and the questions of interest.

3. **Formal data analysis** `[6 MARKS]`

- Appropriate statistical methods have been correctly applied that answers the questions of interest and produce appropriate results.
- Clear and proper explanations of analysis and presentation of key results.
- Appropriate interpretation of the model and how it answers the questions of interest.

4. **Conclusions** `[2 MARKS]`
    - Discussion of key results and validity of conclusions.
    - Discussion of future work/extensions
5. **General layout** `[1 MARK]`
    - Appropriate layout with nicely displayed output and appropriate use of tables and/or figures with sensible labels and cross-referencing/hyperlinks.
    - Overall visual appeal of the report/readability
    - Overall organisation and structure of report is clear

In addition to the final report, you will also be assessed on your R Code:

6. R Code `[3 MARKS]`
    - **Reproducibility** – Code must run error-free and regenerate all report results from start to finish.
    - **Organization & clarity** – Script should be logically structured, well-commented, and match the report's flow.
    - **Technical completeness** – All libraries, data steps, and analyses used in the report must be included and correctly implemented.

2.2.1 Contribution Form

Alongside the group report, each member will be need to separately submit a **Contribution Form** as a **.pdf** file via the **Group Project Contribution Form Submission** link in section **Group project** on the Moodle page. You should submit your contribution form as **GroupNumber_MatriculationNumber_ContributionForm.pdf**. Here, you will provide information on how much you believe each member of your group contributed to the group project. Contribution of each member may consist of the following:

- attending arranged group meetings;
- analysis of the data set via writing `R` code and/or interpretation of the results;
- writing the groups findings and sections of the report; and
- being cooperative and supportive throughout the project.

If a meaningful discrepancy in the contributions of group members is observed then that may result in different grades being awarded to different group members. For example, no grade will be awarded to a group member who does not contribute anything to the group project.

Download data contribution form

2.2.2 Declaration of Originality Form

Each member should also submit a **Declaration of Originality Form** as a **.pdf** file via the **Group Project Declaration of Originality Form Submission** link in section **Group project** on the Moodle page. You should submit your Declaration of Originality form as **GroupNumber_MatriculationNumber_DeclarationForm.pdf**.

Download Declaration of Originality Form

3 Group allocations

Below you can find the information of the classmates you will partner up for the project. Please contact the classmates in your group.

Show 10 entries

Search:

Student Name	GUID	Email	Group
All	All	All	All
Scott Aitken	3178693	3178693A@student.gla.ac.uk	Group 1
Orla Bell	2664735	2664735B@student.gla.ac.uk	Group 1
Katie Colligan	2658553	2658553C@student.gla.ac.uk	Group 1
Kyle Kim	2718740	2718740K@student.gla.ac.uk	Group 1
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Kara Scott	2654584	2654584S@student.gla.ac.uk	Group 2
Shihong Wang	2792750	2792750W@student.gla.ac.uk	Group 2

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4 Group Project descriptions

Air pollution is a significant public health problem. The World Health Organisation estimates that 4.2 million deaths occur every year due to exposure to ambient pollution, with 9 out of 10 people breathing air containing high levels of pollutants. To determine the air quality, concentrations of single air pollutants are of particular interest. Here, we focus on modelling monthly maximum concentrations of a single pollutant in California, US. The raw data contain monthly maximum concentration of different pollutants and measurements of meteorological covariates at different stations across California.

Monthly concentrations of a one pollutant only should be modelled using a EVT framework describing the possible influence of covariates.

> **Important**
>
> Ensure your group downloads and analyses the correct data set you have been assigned otherwise your report will be void and you will not receive a grade.

Projects 1, 2 & 3 - Modelling Sulfur dioxide concentration in California

This group projects will focus on modelling the monthly concentration of Sulfur dioxide measured at specific monitoring stations across 9 different California cities from 2007 to 2016. For some monitoring sites additional covariates such as wind speed, outdoor temperature and barometric pressure are also available.

The data are stored in `group_X_data.csv` and contain the following columns.

- `city` - City where pollutant concentrations where recorded
- `year` - Year when the observation was made
- `month` - Month when the observation was made
- `time` - time index (you can ignore this)
- `long` - longitude for the monitoring station
- `lat` - latitude for the monitoring station
- `site` - name of the monitoring station
- `Wind Speed - Resultant` - Wind speed recorded at the monitoring site
- `Outdoor Temperature` - Temperature recorded at the monitoring site
- `Barometric pressure` - Barometric pressure recorded at the monitoring site
- `Sulfur dioxide` - Monthly Sulfur dioxide concentrations

Questions of interest

The main questions of interest are:

- What constitutes an *extreme' pollution* level for regulatory purposes, and how should such events be identified in monitoring data?
- What can we say about the temporal evolution of extreme pollution levels?
- Is the extreme pollution levels similar across cities?
- What is the effect of the meteorological covariates on the pollutant concentrations?
- Are there ways in which we can improve the estimation?

Download Group 1 Dataset

Download Group 2 Dataset

Download Group 3 Dataset

Projects 4, 5 & 6 - Modelling Carbon monoxide concentration in California

This group projects will focus on modelling the monthly concentration of Carbon monoxide measured at specific monitoring stations across 9 different California cities from 2007 to 2016. For some monitoring sites additional covariates such as wind speed, outdoor temperature and barometric pressure are also available.

The data are stored in `group_X_data.csv` and contain the following columns.

- `city` - City where pollutant concentrations where recorded
- `year` - Year when the observation was made
- `month` - Month when the observation was made
- `time` - time index (you can ignore this)
- `long` - longitude for the monitoring station
- `lat` - latitude for the monitoring station
- `site` - name of the monitoring station
- `Wind Speed - Resultant` - Wind speed recorded at the monitoring site
- `Outdoor Temperature` - Temperature recorded at the monitoring site
- `Barometric pressure` - Barometric pressure recorded at the monitoring site
- `Carbon monoxide` - Monthly Sulfur dioxide concentrations

Questions of interest

The main questions of interest are:

- What constitutes an *extreme' pollution* level for regulatory purposes, and how should such events be identified in monitoring data?

- What can we say about the temporal evolution of extreme pollution levels?
- Is the extreme pollution levels similar across cities?
- What is the effect of the meteorological covariates on the pollutant concentrations?
- Are there ways in which we can improve the estimation?

Download Group 4 Dataset

Download Group 5 Dataset

Download Group 6 Dataset

Projects 7, 8 & 9 - Modelling Carbon Ozone levels in California

This group projects will focus on modelling the monthly levels of Ozone measured at specific monitoring stations across 9 different California cities from 2007 to 2016. For some monitoring sites additional covariates such as wind speed, outdoor temperature and barometric pressure are also available.

The data are stored in `group_X_data.csv` and contain the following columns.

- `city` - City where pollutant concentrations where recorded
- `year` - Year when the observation was made
- `month` - Month when the observation was made
- `time` - time index (you can ignore this)
- `long` - longitude for the monitoring station
- `lat` - latitude for the monitoring station
- `site` - name of the monitoring station
- `Wind Speed - Resultant` - Wind speed recorded at the monitoring site
- `Outdoor Temperature` - Temperature recorded at the monitoring site
- `Barometric pressure` - Barometric pressure recorded at the monitoring site
- `Ozone` - Monthly Ozone levels

Questions of interest

The main questions of interest are:

- What constitutes an *extreme' pollution* level for regulatory purposes, and how should such events be identified in monitoring data?
- What can we say about the temporal evolution of extreme pollution levels?
- Is the extreme pollution levels similar across cities?

- What is the effect of the meteorological covariates on the pollutant concentrations?
- Are there ways in which we can improve the estimation?

Download Group 7 Dataset

Download Group 8 Dataset

Download Group 9 Dataset

Projects 10, 11 & 12 - Modelling Nitric oxide (NO) concentrations in California

This group projects will focus on modelling the monthly Nitric oxide (NO) concentrations measured at specific monitoring stations across 9 different California cities from 2007 to 2016. For some monitoring sites additional covariates such as wind speed, outdoor temperature and barometric pressure are also available.

The data are stored in `group_X_data.csv` and contain the following columns.

- `city` - City where pollutant concentrations where recorded
- `year` - Year when the observation was made
- `month` - Month when the observation was made
- `time` - time index (you can ignore this)
- `long` - longitude for the monitoring station
- `lat` - latitude for the monitoring station
- `site` - name of the monitoring station
- `Wind Speed - Resultant` - Wind speed recorded at the monitoring site
- `Outdoor Temperature` - Temperature recorded at the monitoring site
- `Barometric pressure` - Barometric pressure recorded at the monitoring site
- `Ozone` - Monthly Nitric oxide (NO) concentrations

Questions of interest

The main questions of interest are:

- What constitutes an *extreme' pollution* level for regulatory purposes, and how should such events be identified in monitoring data?
- What can we say about the temporal evolution of extreme pollution levels?
- Is the extreme pollution levels similar across cities?
- What is the effect of the meteorological covariates on the pollutant concentrations?

- Are there ways in which we can improve the estimation?

Download Group 10 Dataset

Download Group 11 Dataset

Download Group 12 Dataset